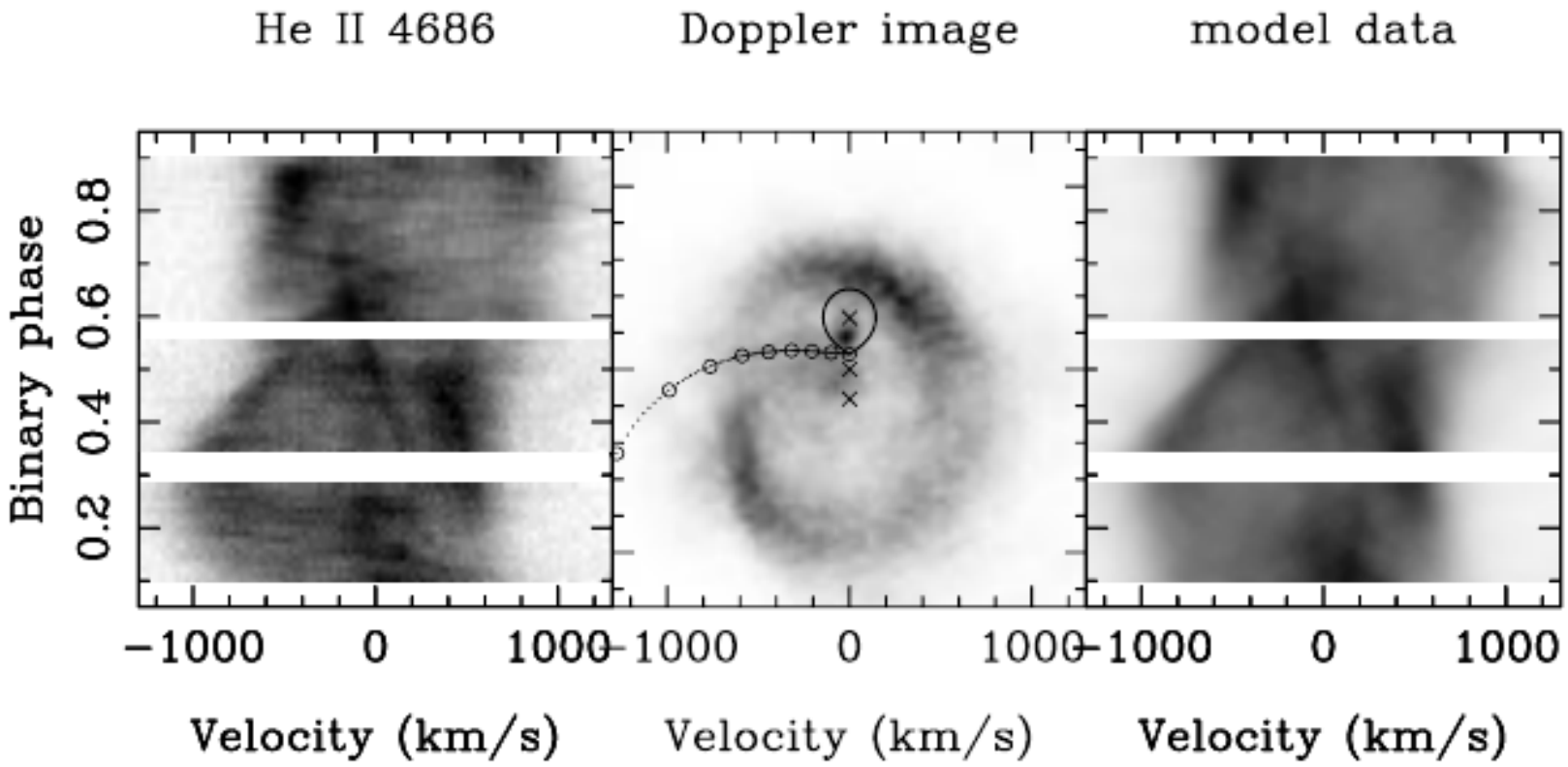
Spiral waves

and their role in angular momentum transfer

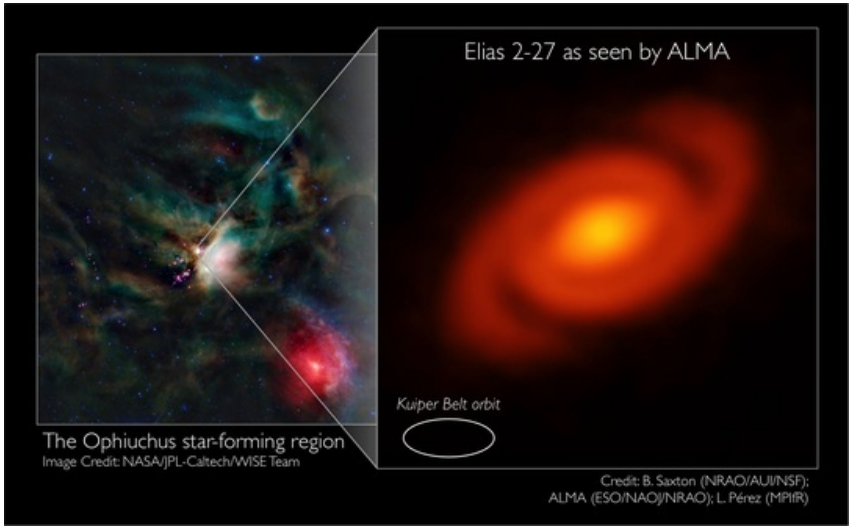
May 4, 2018

CV tomography

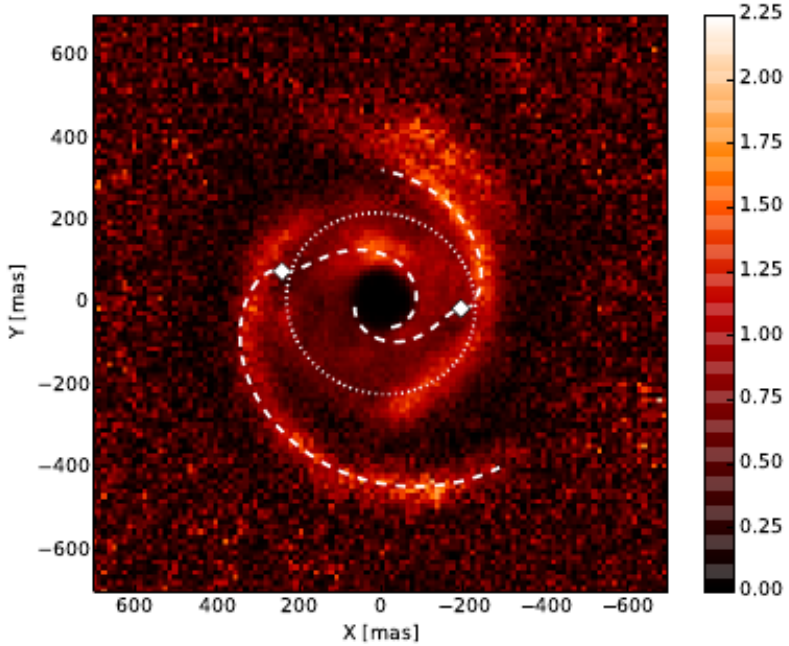


IP Peg (Steeghs et al., 1997; Harlaftis et al., 1999)

Protoplanetary discs



Perez et al.(2016)
ALMA data, CO emission



MWC 758 (Benisty et al., 2015)
NIR (VLT)

Summary of observational data

CV:

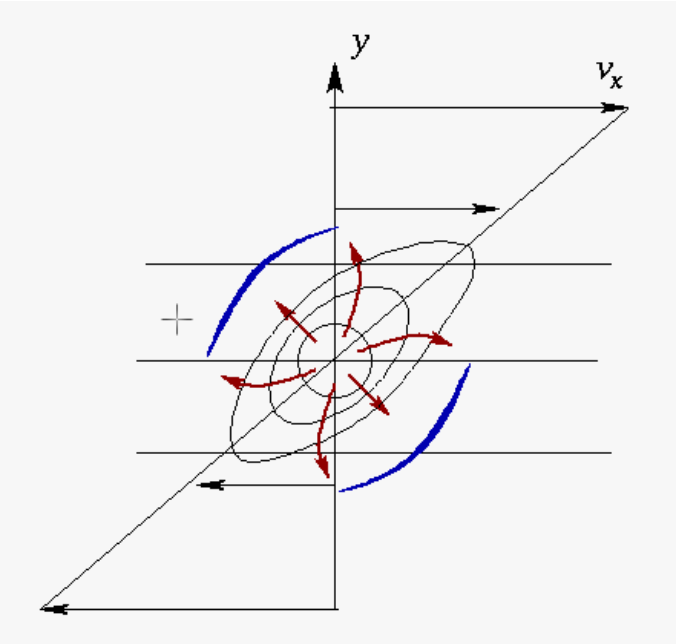
- observed in emission lines (tomography)
- in several objects (IP Peg, ...)
- origin: tidal field of the donor

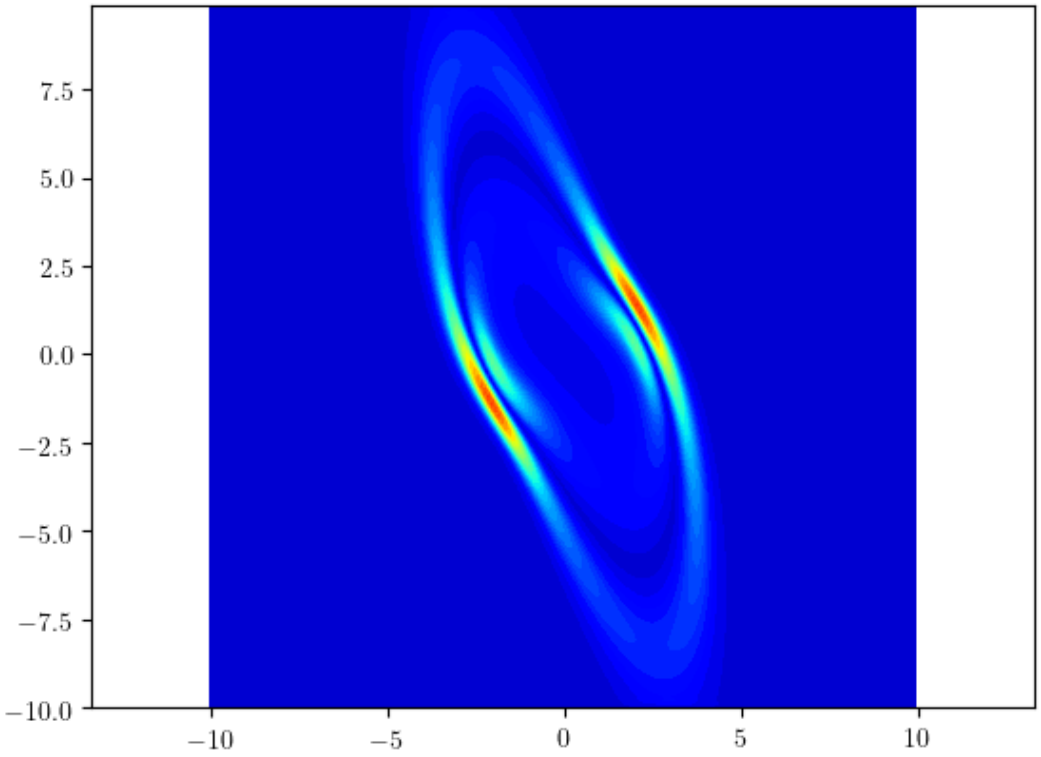
PPD:

- observed in sub-mm (ALMA), and optical/IR (reflection by dust)
- in a large part of all the PPD
- origin: forming planets OR self-gravity OR illumination instability

Why spirals?

Wave front expansion in a constant shear flow:



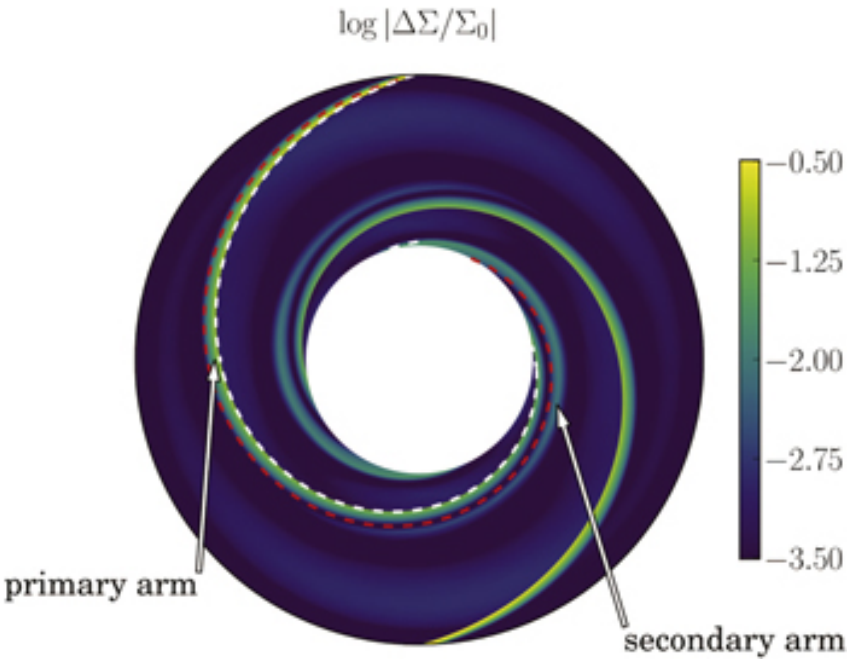


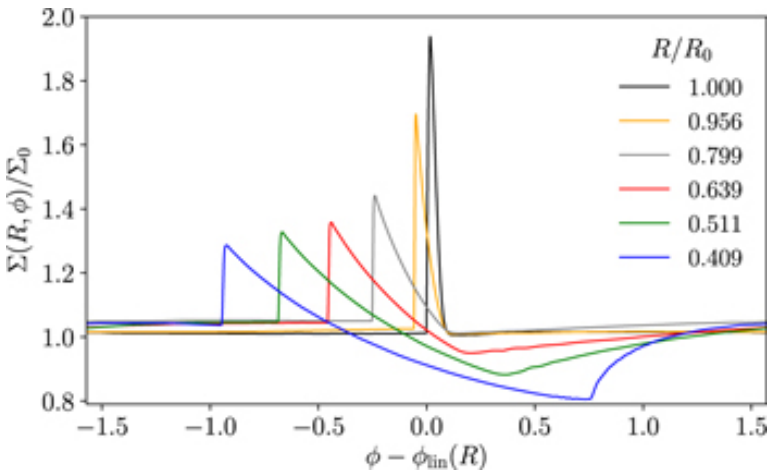
Remember Huygens or Fermat principle

$$\tan\theta \sim \frac{v}{c_s}$$

Their role in dynamics

- develop into shocks ⇒heating, dust formation, radiation etc.
- provide Reynolds stress ⇒effective viscosity and non-local momentum transfer





Arzamasskiy&Rafikov 2017

Reynolds stress

Without any explicit viscous terms

$$\frac{\partial}{\partial t}\left(\Sigma R v_\varphi\right) + \frac{1}{R}\frac{\partial}{\partial R}\left(R^2 \Sigma v_R v_\varphi\right) = 0$$

After averaging over time (stationary flow)

$$-\frac{\dot{M}}{2\pi}\frac{\partial}{\partial R}\left(j\right) = -\frac{\partial}{\partial R}\left(R^2 W_{R\varphi}\right),$$

where

$$j = v_\varphi R$$

$$W_{R\varphi} = \langle \Sigma v_R v_\varphi \rangle$$

Reynolds stress from spiral waves

Turbulent Reynolds stress:

$$\langle \Sigma \delta v_R \delta v_\varphi \rangle = \Sigma \left(\langle \delta v_R^2 \rangle \frac{1}{R} \frac{d}{dR} \left(\Omega R^2 \right) - 2 \langle \delta v_\varphi^2 \rangle \Omega \right)$$

- Normally imcompressible ($\Rightarrow$only $\delta v_R - \delta v_\varphi$ correlation)
- requires some instability process
- works as viscosity for isotropic turbulence

"Stirred not shaken"

Oblique shock:

$$\langle \Sigma \delta v_R \delta v_\varphi \rangle \simeq \Delta\Sigma \Delta v^2 \sin\theta \cos\theta \underbrace{\propto \left(\frac{\Delta\Sigma}{\Sigma} \right)^3}_{\text{for small } \Delta\Sigma}$$

- relies on compressibility
- requires a dynamical perturbation
- correct, 'viscous', sign is ensured by the process of spiral formation

"Shaken not stirred"

Reynolds stress from spiral waves

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Oblique shock:

$$\langle \Sigma \delta v_R \delta v_\varphi \rangle \simeq \Delta\Sigma \Delta v^2 \sin\theta \cos\theta \underbrace{\simeq \Sigma c_s^2 \sim \Pi}_{\text{for large } \Delta\Sigma} \quad \text{(effective } \alpha\text{)}$$

- relies on compressibility
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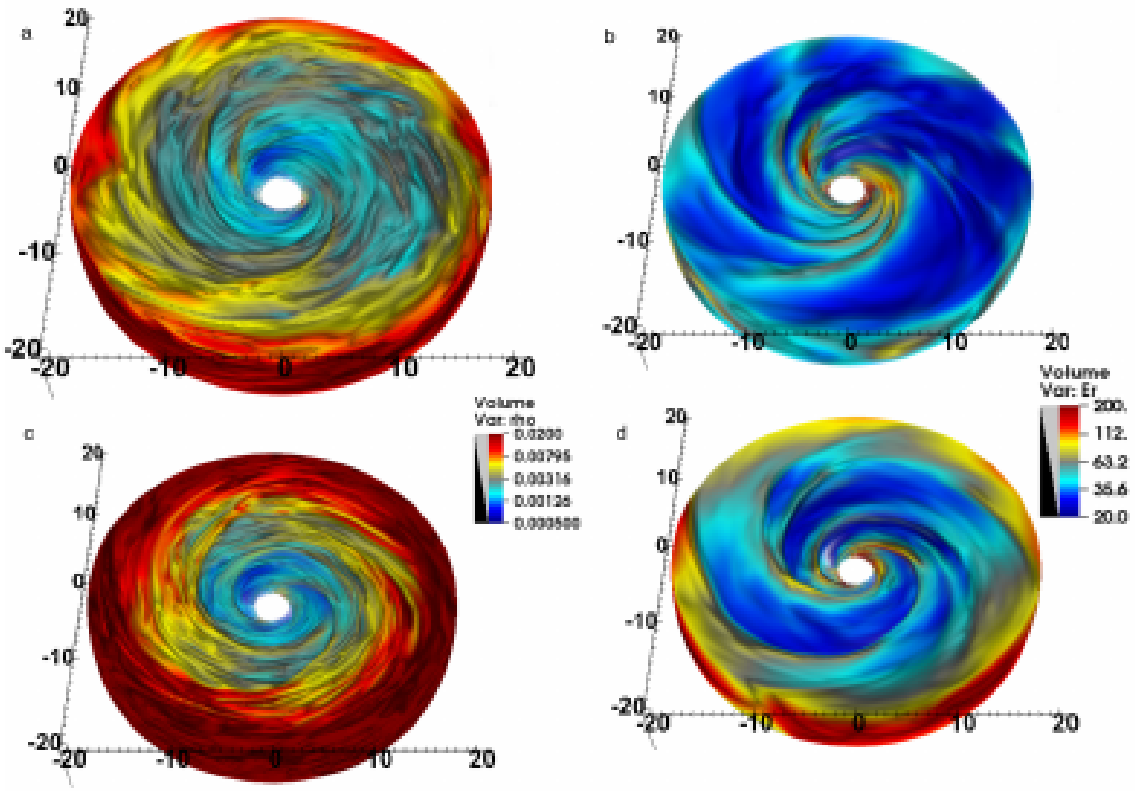
"Shaken not stirred"

Radiation-pressure-dominated case

$$g_z = \frac{\varkappa F}{c} \propto p_{\rm rad} \simeq p$$

⇒pressure does not depend on density
⇒matter does not care about compression, as long as radiation diffusion is rapid



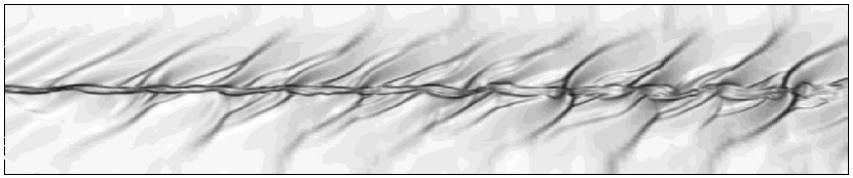
Jian et al. (2017)
Two different models, left panels are density, right are $p_{\rm rad}/p_{\rm g}$ maps.
Reynolds stress dominates

Non-local transfer:

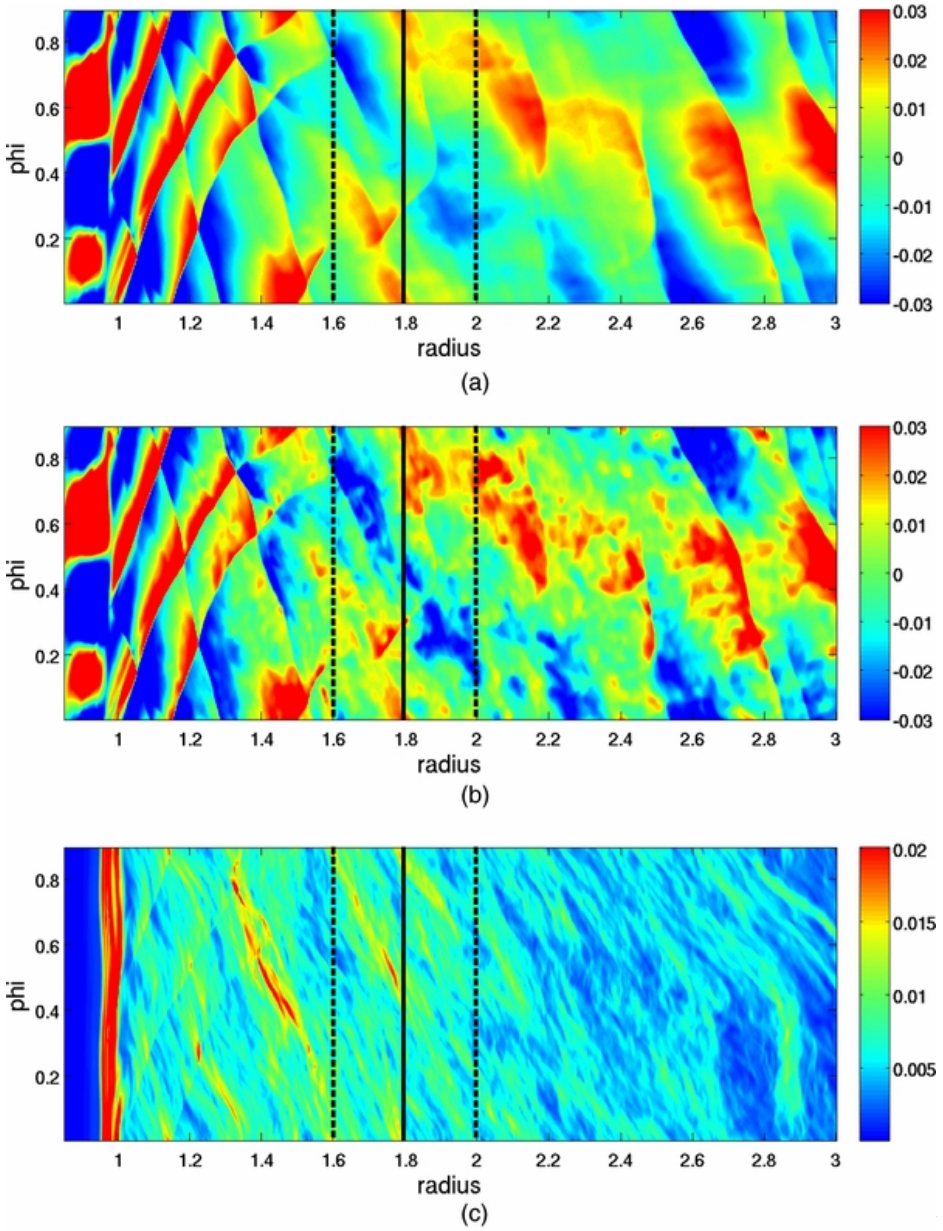
$$W_{r\varphi} \neq \nu\rho R\frac{d\Omega}{dR}$$

Instead, take angular momentum in one place and deposit it in another.

Boundary layer ringing due to KH instability



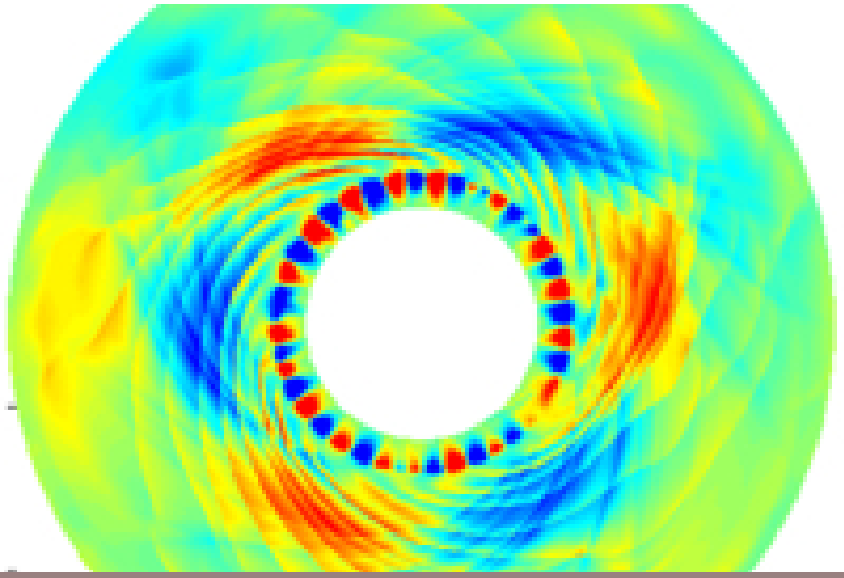
Belyaev et al.(2013) $\rightarrow$



Two cases for spiral shocks

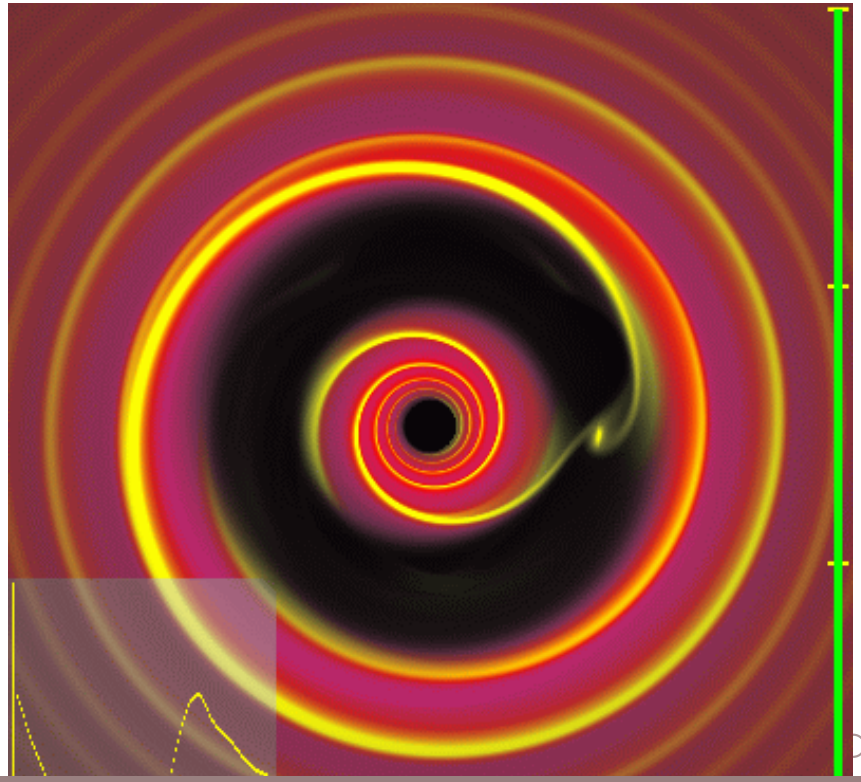
Self-generated:

- local instabilities (radiation pressure, numerous hydrodynamical instabilities)
- boundary layer
- irradiation instability (important for PPD)



Tidally driven:

- donor star
- planets
- self-gravity



Summary and references

- spiral waves are a direct consequence of velocity shear and rotation
- they evolve into shocks if the amplitude is high enough
- they require an energy source: instability or tidal perturbation
- spiral shock waves transport angular momentum (Balbus 2013: "Enhanced angular momentum transport in accretion discs")
- if they do it too efficiently, they can destroy the disc (like in PPD, see Armitage 2015 arXiv:1509.06382: "Physical processes in protoplanetary discs")
- strong multiple spiral shocks reproduce α-like viscosity (but non-local!) and may be responsible for enhanced angular momentum transfer in radiation-pressure-dominated discs (Jiang, Yan-Fei 2017 arXiv:1709.02845: "Super-Eddington Accretion Disks around Supermassive black Holes")
- spiral shock waves generated by shear instabilities are an important constituent of boundary layer accretion (Belyaev et al. 2013 arXiv:1212.0580, 1301.4573)